EXPERIMENT – 1 INTERFACING WITH ARDUINO

PROJECT NAME

Digital Temperature Monitoring System

OBJECTIVE / PROBLEM STATEMENT

The main objective of this project is to design a smart temperature monitoring system using Arduino UNO and a DHT11 temperature sensor.

The system continuously measures room temperature and displays the live temperature value on an SSD1306 OLED display. According to the temperature value, the system automatically controls cooling and heating devices such as a fan and heater.

A push button is also provided to turn the complete system ON or OFF manually.

COMPONENTS USED

Component	Quantity
Arduino UNO	1
DHT11 Temperature Sensor	1
SSD1306 OLED Display	1
Red LED	1
Green LED	1
DC Fan / Motor	1
Heater / Bulb	1
Push Button	1
Jumper Wires	Required

PIN CONNECTIONS

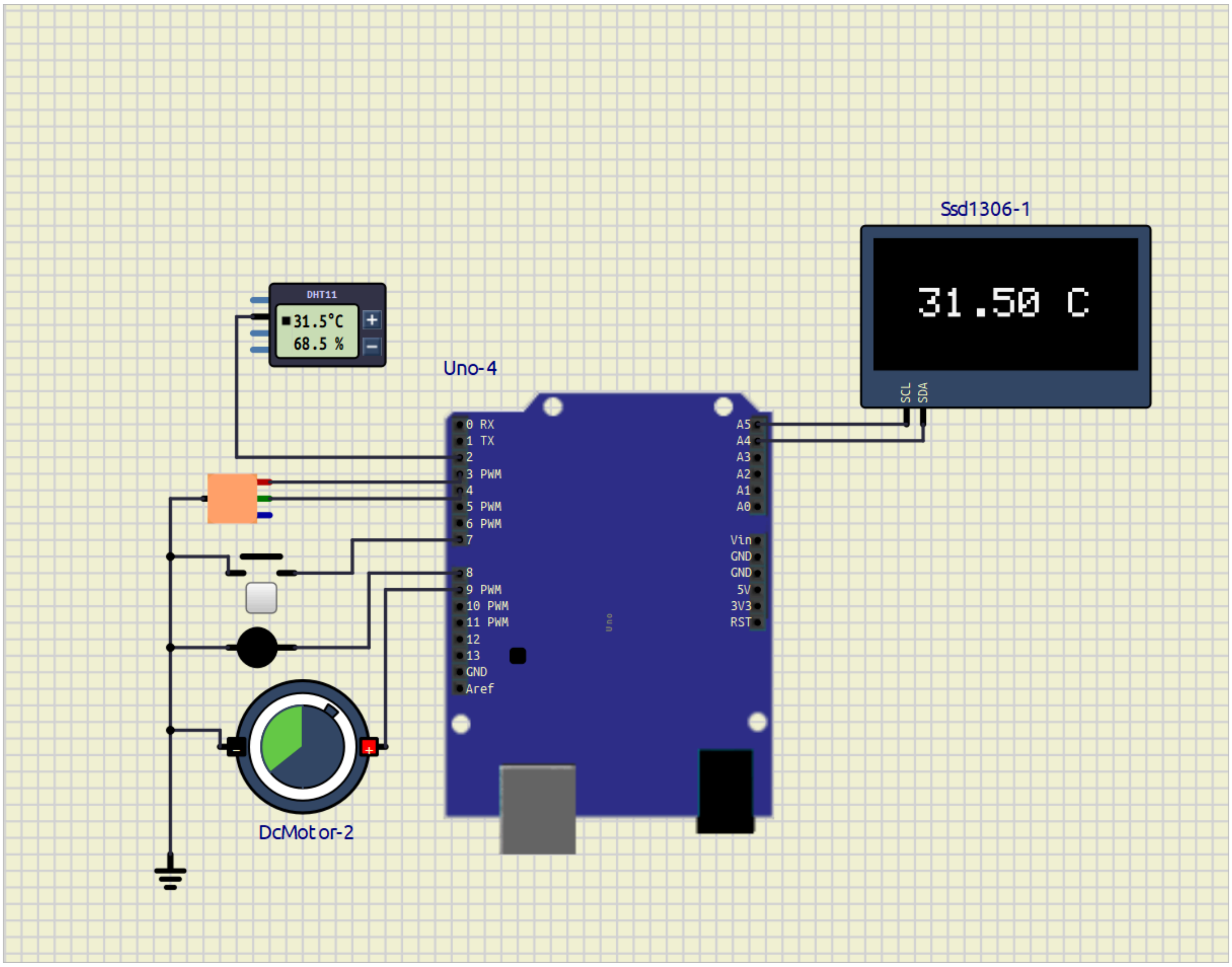
Device	Arduino Pin
DHT11 Data Pin	D2
Red LED	D3
Green LED	D4
Push Button	D7

Device	Arduino Pin
Heater	D8
Fan	D9
OLED SDA	A4
OLED SCL	A5

SOFTWARE USED

- Arduino IDE
- SimulIDE

CIRCUIT DIAGRAM



REQUIRED LIBRARIES

```
#include <Wire.h>
#include <Adafruit_GFX.h>
```

```
#include <Adafruit_SSD1306.h>
#include <DHT.h>
```

MAIN ARDUINO CODE

```
[Code](/Exp-1/smart_temp_controler/temp_sys.ino)
```

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

Adafruit_SSD1306 display(
  SCREEN_WIDTH,
  SCREEN_HEIGHT,
  &Wire,
  -1
);

bool systemON = false;

void setup()
{
  dht.begin();

  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);

  display.clearDisplay();

  pinMode(3, OUTPUT);

  pinMode(4, OUTPUT);

  pinMode(9, OUTPUT);

  pinMode(8, OUTPUT);

  pinMode(7, INPUT_PULLUP);
}
```

```
void loop()
{
  if(digitalRead(7) == LOW)
  {
    systemON = !systemON;

    delay(300);
  }

  if(systemON)
  {
    float temp = dht.readTemperature();

    display.clearDisplay();

    display.setTextSize(2);

    display.setTextColor(WHITE);

    String text = String(temp) + " C";

    int x = (128 - (text.length() * 12)) / 2;

    int y = (64 - 16) / 2;

    display.setCursor(x, y);

    display.print(text);

    display.display();

    if(temp >= 28)
    {
      digitalWrite(3, HIGH);

      digitalWrite(4, LOW);

      digitalWrite(9, HIGH);

      digitalWrite(8, LOW);
    }
    else
    {
      digitalWrite(3, LOW);

      digitalWrite(4, HIGH);

      digitalWrite(9, LOW);

      digitalWrite(8, HIGH);
    }
  }

  else
```

```
    {
      digitalWrite(3, LOW);

      digitalWrite(4, LOW);

      digitalWrite(9, LOW);

      digitalWrite(8, LOW);

      display.clearDisplay();

      display.display();
    }

    delay(1000);
  }
```

WORKING PRINCIPLE

1. The push button is used to turn the complete system ON or OFF.

2. When the system is ON, the DHT11 sensor continuously measures room temperature.

3. The measured temperature value is displayed on the OLED screen in real time.

4. The Arduino UNO compares the measured temperature with the predefined threshold temperature value.

5. If the temperature becomes greater than or equal to 28°C:

   - Red LED turns ON
   - Fan starts running automatically
   - Heater turns OFF

6. If the temperature becomes lower than 28°C:

   - Green LED turns ON
   - Heater turns ON automatically
   - Fan turns OFF

7. The OLED display continuously updates the live temperature reading.

8. When the push button is pressed again:

   - The complete system turns OFF
   - OLED display becomes blank
   - LEDs turn OFF
   - Fan and heater stop working

ADVANTAGES

- Automatic temperature control
- Real-time temperature monitoring
- Easy ON/OFF operation
- Low-cost implementation
- User-friendly system
- Useful for smart automation

APPLICATIONS

- Smart room temperature control
- Automatic cooling systems
- Automatic heating systems
- Industrial temperature monitoring
- Home automation projects
- Embedded systems learning

CONCLUSION

The Digital Temperature Monitoring System using Arduino UNO and DHT11 sensor was successfully implemented and tested in SimulIDE.

The system continuously monitored temperature and automatically controlled the fan and heater according to the temperature value. The OLED display successfully showed live temperature readings, and the push button provided convenient ON/OFF control of the complete system.

This project helped in understanding:

- Arduino interfacing
- Sensor interfacing
- OLED display handling
- Embedded system automation
- Temperature-based control systems

REFERENCES

1. Arduino UNO Documentation
   https://docs.arduino.cc/hardware/uno-rev3/

2. DHT11 Sensor Documentation
   https://components101.com/sensors/dht11-temperature-sensor

3. Adafruit SSD1306 Library
   https://github.com/adafruit/Adafruit_SSD1306

4. Adafruit GFX Library
   https://github.com/adafruit/Adafruit-GFX-Library

5. Arduino DHT Sensor Library
   https://docs.arduino.cc/libraries/dht-sensor-library/

6. Arduino UNO Pinout
   https://deepbluembedded.com/arduino-uno-pinout/